EXT CSCI E-106 Model Data Class Group Project

Analysis of House Prices in King County, Washington

Team 1

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Executive Summary

Our analysis of the open price data for homes in King County, Washington in 2014-2015 focused on finding a model that could predict price while considering multiple important predictors for home value. After exhaustive testing, we developed a neural network model that we deemed our primary choice for predicting house prices. Similarly, we designated a backup model: a transformed model based on an ordinary least squares linear regression model that then had the prices power-transformed to better linearize the data. The benchmark we chose was the lasso regression model of the housing predictors. Taking into account the strengths of these models, we needed to also consider their weaknesses in our analysis. We could not fully address unequal variance in the predictors as well as normality of the residuals in our backup model. We would also raise that interpreting the neural network models themselves may present difficulty due to the nature of neural networks themselves. In spite of these flaws, they are our best models.

I. Introduction

The purpose of this project is to develop a model that can predict home sales prices utilizing the data derived from home sales in King County, Washington and finding what predictors are most significant to this model. By doing so, we gain a greater understanding of the King County real estate market. Our dataset is an open-source compilation of house sales from King County taken from May 2014 to May 2015. This dataset is composed of 21613 entries, each with 21 variables, that were then split into training and test data sets made up randomly selected entries from the original data in a 70:30 split. Specifically, the test set is made up of 6484 randomly selected entries while the model was trained on a dataset made up of 15129 randomly selected entries - all without replacement.

After we finished preprocessing the data, we tested a variety of regression models in R to find an optimal model for predicting house prices. We tested and transformed models including first order least squares regression models, stepwise regression to have only statistically significant variables in the model, robust regression to address outlying data points, and explored logistic regression. In addition, we utilized ridge regression, lasso regression, elastic net regression to address multicollinearity. Finally, we made use of alternative, more advanced methods of modeling including regression trees and neural networks to help to develop models for predicting responses. We adjusted our models and data depending on whether or not issues such as outliers, nonlinearity, high multicollinearity, and heteroscedasticity were discovered. Our primary means of comparing the efficacy of each model was based on comparisons of the adjusted R-squared when running the trained model with the test data. After an exhaustive investigation of different models, we selected the benchmark and best models.

II. Description of the Data and Quality

Prior to developing any models, we needed to prepare, pre-process, and evaluate our data such that they could be used for modeling. Once we'd taken time to properly understand what variables we had been provided (Figure 1.1, Atchison et. al), our group recognized a need to extensively pre-process, clean, transform and even exclude some of the variables for accurate predictive analysis.

Data Preprocessing

Variable Transformation

Among the first necessary transformations was converting our prices, which were written as strings with dollar signs and decimal points, to a readable numeric format. Similar numeric conversions were done with other variables to ensure that the data was stored in an appropriate numeric format - for example, the “sqft_living” and “sqft_lot” variables.

As the time of year that houses are sold can affect what homes are sold and their prices, we aimed to include the date in our analyses. To do so, we opted to convert the “date” variable, which was initially read as a string, to a numeric format. This was done by utilizing the “lubridate” package to extract various date-related features.

Dummy Variable Creation

By converting categorical data like “waterfront”, “view”, “condition”, and “grade” into factors, we made dummy variables such that we could numerically represent each level of categorical data in our analysis. This meant creating dummy variables for each level of “condition”, “grade”, and “view” - ordinal variables with intrinsic scoring as an evaluation of certain properties or features of the home. Similarly, this also enabled binary data, such as “waterfront” to be represented with a 0 and 1 to show that the home wasn’t or was a waterfront, respectively.

Distance Calculations and Convergence

The Haversine distance is a method for calculating the shortest distance between two points on the surface of a sphere, given their longitudes and latitudes. This is crucial in our

context since the Earth is approximately spherical. The formula accounts for the curvature of the Earth and provides a more accurate distance than a simple straight-line (Euclidean) distance. The Haversine distance (d) is calculated using the following formula in the provided code:

$$\mathbf{d = R \times c}$$

where R is the Earth's radius (approximately 6371 kilometers), and c is the angular distance in radians, computed from the latitude and longitude of the two points.

In our model, the Haversine distance is used to calculate the distance of each property from a specific convergence point. This distance is an essential feature, as it captures the geographical desirability or connectivity of a property, which is a significant factor in real estate pricing.

The convergence point is a calculated geographical point that represents a central location among high-value homes. It's determined by calculating the mean latitude and longitude of homes that fall within the top 5% of prices in the dataset (as defined by the high-value threshold). The convergence point serves as a benchmark location, providing a reference for the relative position of each home in the dataset. By calculating the distance of each property from this point using the Haversine formula, the model can incorporate the aspect of location desirability, which often correlates with higher property prices.

The use of both Haversine distance and the convergence point in tandem enhances the model's predictive power. The Haversine distance provides a precise measure of each property's location relative to a central, high-value area, encapsulating factors like neighborhood desirability and accessibility. Real estate prices are significantly influenced by location. By

quantifying a property's proximity to a desirable location (convergence point), the model gains a nuanced understanding of geographical value. This is especially relevant in areas like King County, where location plays a crucial role in property valuation.

Incorporating geographical features such as the Haversine distance and a specifically calculated convergence point significantly enhances the model's ability to predict housing prices accurately. By accounting for the spherical nature of the Earth and the centrality of location in real estate valuation, these features provide a robust foundation for predictive analysis in the King County housing market. This detailed, data-driven approach, rooted in geographical analysis, positions the model to deliver nuanced and accurate predictions, essential for stakeholders in the real estate sector.

Exclusion and Special Cases

In one instance, we elected to specifically remove a variable from our analysis. Specifically, we removed the variable "id" as its values represented unique identifiers for the sales of the homes - arbitrary values which would not be relevant for our analysis of the prices of homes.

In other cases, we elected to keep certain values that could be alternatively interpreted. For example, the "bathrooms" variable could be measured as the quantity of bathrooms on the property. However, some homes have bathrooms that only have toilets and don't have a full bath or shower in them. These particular bathrooms are represented as halves or 0.5. In these cases, though the variable was measured in two different ways (full baths and half baths), we elected not to modify the variable or the values to retain important details about the homes' characteristics.

Exploratory Data Analysis (EDA)

We conducted an exploratory evaluation of our variables to determine what trends, abnormalities, and qualities could be found while utilizing multiple variables. This would be a necessary step to determine what further changes to our data needed to occur for developing a robust model for price prediction. The types of evaluations we conducted include analysis of continuous variables, categorical variables, correlations, temporal trends, and determining how geographical data played a role in predicting prices.

Continuous Variable Analysis

To gather information on any abnormalities or trends within continuous variables, we conducted analyses such as plotting continuous variables against each other for detection of any trends along with visualizing histograms and boxplots to understand their distribution and detect outliers and skewness within our data. Among the variables tested include “price”, “sqft_living”, “sqft_lot”, “sqft_above”

Among trends detected were great right-skewness in a number of histograms. This includes strong right-skewness in variables such as “sqft_living” (Figure 5.2, Atchison et. al), “sqft_lot” (Figure 5.4, Atchison et. al), “sqft_above” (Figure 5.6, Atchison et. al) and “sqft_basement” (Figure 5.8, Atchison et. al). In this context, this means that a majority of homes are distributed on the left sides of the histogram or have fewer points with greater sizes or distances on the right side.

Individually, we see that the living space histograms are right-skewed - this means that most homes have smaller living spaces while fewer homes have very large living spaces.

Similarly, the histogram of lot size is right skewed as well, indicating that very few homes have large lot sizes - the majority have relatively smaller lots or none at all. In the square feet above ground histogram, we noted that more homes had relatively smaller spaces above ground with few having large spaces above ground. The basement size histogram showed that most homes have little or even no basement space within them.

Categorical Variable Analysis

During our categorical variable analysis, we visualized and evaluated the distribution, count, and appearance of our categorical variables like “bedrooms”, “bathrooms”, “floors”, “waterfront”, “view”, “condition”, and “grade” using bar charts and boxplots. This lets us understand not only how each variable but also how each level of the variable may significantly influence prices or may not have any impact at all. It also lets us examine frequencies of each factor in our data. One detail we did note was that having a waterfront as opposed to not saw a great increase in the median price of the homes (Figure 5.21, Atchison et. al). Utilizing our categorical data, we are able to corroborate that the presence of a “waterfront” is a significant predictor of house price.

Correlation Analysis

In our correlation analysis of the King County housing market we are trying to understand how continuous variables correlate with each other and, more importantly, with the target variable price. Illustrated in Table 5.1 (Atchison et. al) we’ve showcased the top 20 correlation values concerning the target variable, price, with the values sorted by their absolute magnitudes. Within our observation from this analysis, we were able to witness positive

correlation with price. Variables such as sqft_living, sqft_above, sqft_living15 and bathrooms exhibit robust positive correlations with the target variable (price). This implies that as features increase, house prices tend to increase correspondingly. Features such as grade_Above_Average and view_4 also demonstrate positive correlations, indicating that properties with higher grades and better views tend to command higher prices.

The strength of these correlations provides insights into the importance of variables in predicting house prices and variables such as sqft_living and sqft_above emerge as strong predictors of price.

Temporal Trend Analysis

For our temporal trend analysis, we focused on analyzing the influence of time-related features such as month and season on the house prices. For example, we evaluated monthly trends in average house prices, seasonal trends in average house prices, week of the year trends in the average house prices, and day of the year trends in average house prices.

The “Daily Trends in Average House Prices" line graph (Figure 5.36, Atchison et. al) showcases the average house prices for each day of the year. By plotting these data, we can identify patterns and potential price variations that may be influenced by specific dates or events.

Geographic Influence Analysis

In our collaborative analysis of the King County housing market, illustrated in Figure 5.38, "Distance to Convergence Map," we have identified a significant relationship between home prices and their proximity to a centrally located convergence point (Atchison et. al). This point, highlighted by the innermost ring on the map, represents the geographic average of the top

5% of home prices and serves as the origin for 2 km concentric distance rings that radiate outward. The color-coded z-score intervals on the map—ranging from green for average-priced homes to red for the most expensive—reveal a discernible pattern: the closer a home is to the convergence point, the higher its price tends to be.

The heatmap clearly shows clusters of high-value properties (in red and orange) surrounding the convergence point, suggesting that these areas are highly sought after, likely due to favorable amenities, school districts, or other desirable features that tend to drive up home values. As the distance from the convergence point increases, we observe a gradual transition to homes priced closer to the county average, indicated by the green-colored areas. These visual insights have been pivotal in refining our predictive model, enabling us to quantify the impact of location on home prices with greater accuracy. We acknowledge that the desirability of a location, as proxied by the convergence point, is a potent determinant of home prices in King County. Our model, thus, incorporates this spatial dynamic, considering both the distance from the convergence point and the distribution of prices as crucial variables. Moving forward, we aim to enhance the model by integrating additional layers of data, thereby improving our capacity to forecast housing prices with even higher fidelity.

III. Model Development Process

In this section we'll give more details on how we developed the models, addressed linearity assumptions with transformation and remedial measures.

OLS Model Development and Stepwise Regression

Our first step in the model development process was to utilize our cleaned and transformed training data to develop an Ordinary Least Squares (OLS) model for predicting house prices. We utilized the R-squared and the adjusted R-squared to evaluate the strength of the fit of our model. Our initial ordinary least squares model had an R-squared of 0.8186 and an adjusted R-squared of 0.8174 (Section 4.3, Atchison et. al). This means that, after cleaning and transformation of our data and fitting the training data to a linear model, approximately 81.74% of the variability in house prices could be explained by our model when adjusted for the number of predictors. Due to how close the R-squared is to 1, we surmised that we had a strong fit with our linear model but that there was potential for further improvement of our model.

Using our preprocessed data for the linear models, we made new ordinary least squares linear models with all variables for backwards, forwards, and best possible stepwise regression models to add or remove variables from while searching for a model with all significant coefficients.

During the analysis, we came to the conclusion that of the models, the stepwise model that both added and removed features based on significance was a potentially valuable and powerful one due to its methodology of presenting only the most significant predictors. As we saw from our list of dropped features from each model (Table 6.1, Atchison et. al), OLS_Step_Both had the most dropped features of both models. With this, we maintain predictive power while also finding the most influential variables.

The data showed that the OLS linear model and all stepwise models had around the same adjusted R-squared for the training data. About 81.61-81.86% of all variability could be

explained by all of the models (Table 6.2, Atchison et. al). All of our stepwise models had a marginally weaker performance compared to the ordinary least squares linear model.

Looking at the predictors for the ordinary least squares regression model, we found that at least one variable was insignificant: specifically, “sqft_lot15” had a p-value larger than 0.05. On the other hand, “sqft_living”, “waterfront1” (which corresponds to the home being a waterfront property), and “bedrooms” all had significant coefficients. From these insights, we came to the conclusion that they had minor and major effects on prices, respectively. Finally, we detected a considerably large spread in residual values which could suggest potential non-linear relationships that weren’t captured by our model or outliers that impact our model. To check for these, we made diagnostic plots to evaluate for outliers, heteroscedasticity, and nonlinearity.

Diagnostic Plots (OLS)

Linearity assumption

We began our diagnostic evaluation of our ordinary least squares linear model by visually checking for linearity in our model. As it is shown (Figure 7.1, Atchison et. al), our red Lowess fit line in our residuals vs fitted values plot is an almost entirely straight line that coincides with our horizontal axis. Furthermore, until some of the highest predictor values, we maintain a relatively even and tightly confined spread of residuals along the lowess line. We see there are at least a few outliers with higher than normal residual values from the training data (13244, 9986, 5127).

Normality of Residuals

Next, we evaluated a Q-Q plot to visually check the normality of our model's residuals (Figure 7.4, Atchison et. al). When evaluating the Q-Q plot, we saw that our distribution very closely followed the line along the middle of the Q-Q plot but had differences in their deviation on each end of the plot. On the left side of our Q-Q plot, slight deviation from normality occurs without outliers to note. On the other hand, deviation from our normal distribution happens earlier on the right side and 3 outliers that may be affecting our distribution were noted (5127, 9986, 13244). Going by our plot, our residuals are mostly normally distributed with greater deviation as our predictors and dependent variables increase in value.

Constant Variance Assumption

To visually check our model for heteroscedasticity, we evaluated a scale-location plot (Figure 7.5, Atchison et. al). Looking at the plot, we see that the spread of standardized residuals around the average of those residuals is relatively equal. The shape of the spread does not narrow or widen or take any other shape. Though the spread of residuals does thin out at higher fitted values, the evenness of the spread does not change. This suggests that our model has an equal variance in its residuals.

Outliers

Finally, we checked Cook's distance to visually determine whether or not there were any influential outliers that may have impacted our model (Figure 7.6-7.7, Atchison et. al). Our plots indicate that the data with the highest Cook's distance were points 5127, 9986, and 13244 - these

were potential outliers. When comparing them to our threshold, 5127, 9986, and 13244 had greater Cook's distance than our threshold of 4/n (n = quantity of data points in training data). These two points were found to be the most influential of outliers within our model. The most significant outlier of them was 13244 with a Cook's distance of nearly 0.30.

To improve our model, we chose to filter out data that had a higher Cook's distance than 0.04 and remove the influential outliers from our training and test data. After having removed these outliers, we remade our ordinary least squares linear model and found that our new adjusted R-squared was 0.8208. The fit of our best linear model had improved with the removal of these data.

Addressing Linearity, Multicollinearity, Normality and Heteroscedasticity

Nonlinearity

We considered that our distribution may benefit from a non-linear transformation to better capture the best fitting model. Therefore, we conducted a Box-Cox transformation to improve our predictive power and linearize our regression. We were able to identify the lambda at which maximum log-likelihood was achieved and found that the lambda in question was 0.0909091 (Figure 8.1, Atchison et. al).

From there, we implemented a power transformation of our response variable "price" for our new "Transformed Model". When examining it, we found that all predictors were significant. We also obtained an R-squared of 0.8726 and an adjusted R-squared of 0.8719 - our highest up till this point. This was the best fit model we'd had till now.

When we evaluated our diagnostic plots of our transformed model, our residuals vs fitted and scale-location plots' spreads hadn't changed much. Yet, new outliers could be seen and upon evaluation of the Q-Q plot, deviation from normality was detected about as soon as before. Outliers could be seen in the lower or left tail of the Q-Q plot this time - the outliers were 3098, 14109, and 7469. In the Residuals vs Leverage plot, none of the outliers appear to cross Cook's distance threshold (Figure 8.2, Atchison et. al).

Multicollinearity

With our transformed model, we conducted a Variance Inflation Factor-based analysis to determine whether multicollinearity was present amongst our variables. Our threshold for multicollinearity for this analysis was a VIF of 10 to identify predictors with high multicollinearity that may cause instability in the coefficients. From the output of VIF analysis, we did not detect any predictors in our transformed model with issues of multicollinearity - none of them had VIF scores that were greater than our threshold. The highest VIF reported was for our "sqft_living" predictor and it was about 5.59 (Section 8.3.1, Atchison et. al).

Heteroscedasticity

For our analysis into the model's heteroscedasticity, we conducted a Breusch-Pagan test on our transformed model to numerically test for equal variance in the residuals. If we were to reject the null hypothesis that the variance of the residuals was equal across the different levels of the predictors, then we could conclude heteroscedasticity was present.

Our p-value for our Breusch-Pagan test on our transformed model was 2.2e-16, which was less than our alpha value of 0.05 (Section 8.4, Atchison et. al). Therefore, we rejected the null hypothesis and said that the variance in the residuals was not equal among the levels of the

predictors. The variability of error was not constant across all levels of the predictor which may impact the model's reliability.

Remedial Measures

After performing the initial Breusch-Pagan test to assess heteroscedasticity on the weighted linear model (transformed_model_weighted), the test yielded a p-value less than 2.2e-16, indicating significant heteroscedasticity in the model. This initial result indicated that the residuals of the model exhibited a pattern that violated the assumption of constant variance.

To address this issue, 200 iterations were carried out, where the model was repeatedly re-fitted and the Breusch-Pagan test was performed after each iteration. After these iterations, the Breusch-Pagan test still resulted in a p-value still less than 2.2e-16 (Section 8.5.2, Atchison et. al). Despite the numerous iterations, the p-value remained extremely low, indicating that heteroscedasticity persisted in the model.

Even after 200 iterations of weighting the linear model in an attempt to address heteroscedasticity, the model continued to exhibit significant heteroscedasticity, as indicated by the persistently low p-value from the Breusch-Pagan test. Further investigation or alternative modeling approaches may be necessary to effectively address this issue.

We moved forward to check for multicollinearity. We chose ridge, lasso and elastic net regression models to address multicollinearity amongst the variables.We then performed the Huber's method for the robust regression. We utilized robust regression to address influential observations or outliers in the data. Finally, we performed the comparison of coefficients across the models to help us to define the benchmark model.

The coefficient analysis compares various regression models based on the total sum of absolute differences in their coefficients when compared to a set of reference models: “Huber Model,” “Ridge Model,” “Lasso Model,” and “EN Model.” This analysis helps determine which model is most consistent with the reference models in terms of coefficient values, which can be indicative of their effectiveness in addressing diagnostic issues like multicollinearity, outliers, and model stability.

IV. Model Performance Testing

OLS Linear Model

The OLS Linear Model displays a commendable fit to the data with R-squared values of 0.81859 for training and 0.81966 for testing data. These values suggest that the model explains approximately 82% of the variance in housing prices, both in training and testing datasets, indicating a robust model that generalizes well to new data. The similarity in R-squared values between the training and testing sets is particularly noteworthy as it implies a consistent model performance without overfitting.

Regarding error metrics, the RMSE values of 154,041.04 for training and 161,166.07 for testing data suggest a moderate prediction error. While not as low as some more complex models like the Neural Network, these RMSE values are relatively acceptable for a linear regression model. The consistent RMSE between training and testing also underscores the model's generalizability.

The MAE values, 91,957.68 for training and 93,048.1 for testing data, further reinforce this point. They reflect a stable average prediction error across both datasets, which is a positive aspect for model reliability.

The OLS Linear Model stands out for its simplicity and consistency, making it a valuable model for predicting housing prices, especially when interpretability and understanding of linear relationships are crucial.

OLS Step Forward Model

The OLS Step Forward Model, which iteratively adds the most significant variables, shows good model performance with R-squared values of 0.8178 for training and 0.81961 for testing data. These figures indicate that the model explains over 81% of the variance in the housing prices for both datasets, showcasing a balanced fit and good predictive power.

The RMSE values of 154,375.23 for training and 161,191.81 for testing indicate a moderate level of prediction error. These figures, while slightly higher than the OLS Linear Model, are still reasonable for a regression model. The consistency of RMSE between training and testing suggests that the model is not overfitted to the training data and can generalize well.

The MAE values of 92,195.59 for training and 93,107.59 for testing data further confirm the model's stable predictive performance across the two datasets. The similarity in these values suggests that the model's average prediction error is consistent, which is advantageous for practical applications.

The OLS Step Forward Model is a robust option for predicting housing prices, providing a good balance between model complexity and performance. Its incremental approach in selecting variables can be particularly useful in identifying the most relevant features affecting housing prices.

OLS Step Both Model

The OLS Step Both Model, which considers both adding and removing variables in each step, shows R-squared values of 0.81614 for training and 0.81715 for testing data. These values, though slightly lower than the previous OLS models, still indicate that the model explains a substantial portion of the variance in housing prices, with over 81% in both training and testing data.

The model's RMSE values of 155,079.14 for training and 162,285.16 for testing data are somewhat higher than the other OLS models, suggesting a slightly increased average prediction error. However, these values are still within a reasonable range for regression models.

The MAE values of 92,978.7 for training and 94,247.4 for testing data are consistent and reflect the model's stable performance in predicting housing prices across different datasets. This stability is crucial in ensuring the model's reliability in various scenarios.

The OLS Step Both Model provides a balanced approach to variable selection, offering a good mix of predictability and interpretability. While it may not be the strongest model in terms of minimizing error metrics, its comprehensive approach to feature selection makes it a valuable tool in understanding the factors influencing housing prices in King County.

OLS Step Both with Outliers Model

The OLS Step Both with Outliers Model, which presumably includes a stepwise approach while also addressing outliers, shows an R-squared of 0.82176 for training and 0.8167 for testing data. This indicates a strong ability to explain the variance in housing prices, slightly

better than some other OLS models. The slight decrease in R-squared from training to testing could suggest minor overfitting or variance in the test data not captured by the model.

In terms of error metrics, the RMSE values are 148,908.82 for training and 162,486.45 for testing. The lower RMSE in the training set compared to the testing set might be due to the model being slightly more tuned to the training data, especially in the context of outlier handling. However, these values are relatively lower compared to other models, which is a positive sign.

The MAE values of 91,385.31 for training and 93,114.76 for testing are consistent and indicate a stable average prediction error across both datasets. This stability is a positive indicator of the model's reliability across different scenarios.

The OLS Step Both with Outliers Model appears to be a strong contender in terms of balancing the trade-off between complexity and prediction accuracy, especially with its focus on outlier handling, which is crucial in real estate data with its often skewed and outlier-prone distributions.

Transformed Model

The Transformed Model, likely involving some form of data transformation or feature engineering, exhibits strong performance, with R-squared values of 0.87256 and 0.87105 for the training and test datasets, respectively (Table 8.2, Atchison et. al). This consistency in R-squared values between training and testing suggests the model generalizes well without significant overfitting. The RMSE values, standing at 133,184.41 for training and 200,676.10 for testing, are relatively high, particularly in the test set. This indicates a larger average prediction error, which might be a concern in practical applications where lower errors are desired. The MAE values of

76,550.98 for training and 81,099.88 for testing further suggest that, on average, the model's predictions are off by these amounts from the actual prices. The closer values of MAE between training and testing datasets indicate a consistent performance of the model across both datasets. In addition, the MAE values of 76,550.98 for training and 81,099.88 for testing indicate that, on average, the model's predictions deviate by these amounts from the actual prices. The consistent performance of the model across both training and testing datasets, as reflected in similar MAE values, suggests reliability in real-world applications.

Weighted Transformed Model 200 Iterations

The Weighted Transformed Model 200 Iterations, potentially a more complex or specialized model, shows R-squared values of 0.75884 for training and 0.74016 for testing data. These values are comparatively lower than both the linear and the Regression Tree models, suggesting a lesser ability to explain the variance in housing prices. The consistency between training and testing R-squared values, however, does indicate a level of robustness in the model's predictive ability across different datasets.

The RMSE values are significantly higher, with 642,962.63 for training and 662,703.16 for testing, indicating very high average prediction errors. This might raise concerns about the model's practical application, as such high errors could significantly impact decision-making in real estate markets.

Similarly, the MAE values of 537,582.19 for training and 543,270.39 for testing are extremely high, further emphasizing the model's limited accuracy in predicting housing prices. This could be a result of the model's complexity, potential overfitting, or it not being well-tuned to this specific dataset.

The Weighted Transformed Model 200 Iterations Model, despite its potential complexity and sophistication, does not seem to perform as well as the other models in predicting housing prices in King County. Its high error metrics and lower R-squared values suggest that it might not be the best choice for this particular application, especially when accuracy and reliability are paramount.

V. Challenger Models

In undertaking the comprehensive task of building and evaluating an alternative model to predict housing prices in King County, our team engaged in a thorough and detailed process, focusing on a Neural Network (NN) model. This decision was made after considering other approaches like regression trees and exploring logistic regression.

Validating Neural Networks

The selection of a Neural Network (NN) model over alternatives such as regression trees or support vector machines (SVMs) is grounded in the recognition of the NN's unmatched ability to capture intricate, non-linear relationships within the housing price data. Housing markets are inherently complex, influenced by numerous factors that interact in nuanced ways. NNs excel in modeling such complex systems due to their capacity to learn hierarchical representations of data.

We opted for Keras, a high-level neural networks API, to maintain our commitment to leveraging the latest and most versatile tools in the field of deep learning. Keras provides a

seamless interface to the TensorFlow and Theano libraries, offering extensive functionality and flexibility for designing and training neural networks. Its support for fine-grained hyperparameter tuning, coupled with its intuitive architecture for constructing complex models with multiple layers and units, positions it as the ideal choice for crafting a sophisticated NN model.

For one, the normalization of data, achieved through the normalize_data function, represents a fundamental step in preparing data for neural network modeling. Neural networks are highly sensitive to the scale of input features. By scaling each feature to a range between 0 and 1, we ensure that all features contribute equally to the learning process. This prevents features with larger scales from dominating the model's weights and gradients, facilitating convergence and robust training.

Another point in its favor is that the sophistication of the model architecture, constructed using Keras, reflects a deep understanding of the data's complexity. The choice of a sequential neural network with multiple dense layers showcases the intention to capture intricate patterns within the data. The varying number of units (ranging from 32 to 128 units) within these layers provides the model with the flexibility to learn and represent complex, hierarchical relationships.

Employing a deep architecture with multiple layers (256, 128, 64, 32, 16, 8, 4, and 2 units) is a strategically sound decision. It enables the model to capture both high-level and subtle features of the dataset. Deep layers are especially advantageous when predicting housing prices, as real estate markets are influenced by a multitude of factors, each contributing differently to pricing dynamics. These deep layers empower the model to learn and represent these multifaceted relationships effectively.

The systematic exploration of various batch sizes (ranging from 50 to 130) and epochs (ranging from 50 to 250) demonstrates a commitment to optimizing model training. Smaller batch sizes are known to promote generalization by introducing more randomness into the training process, while a wider range of epochs ensures that the model iterates over the entire dataset multiple times, potentially enhancing convergence and learning.

The use of a validation set for monitoring model performance during training is a best practice. It safeguards against overfitting and ensures that the selected model configuration is based on the lowest validation mean absolute error (MAE). This systematic approach aligns with the principle of rigor and thoroughness in model selection.

The meticulous verification of the data's adherence to Neural Network assumptions, including data normalization and addressing multicollinearity through dropout layers, underscores the dedication to model reliability and integrity. In addition, evaluating the model's performance on both training (in-sample) and test (out-of-sample) datasets is a gold standard in model validation. This approach ensures that the model not only fits the training data well but also generalizes effectively to unseen data, which is crucial for real-world predictive applications.

The inclusion of backtesting, involving tests on various subsets of data, enhances the robustness assessment of the model. It provides valuable insights into how the model performs under different market conditions and over varying time periods, contributing to a comprehensive understanding of its stability and consistency.

The incorporation of a rescaling function for converting normalized predictions back to their original scale is essential for practical interpretation and decision-making. It enables stakeholders to compare predicted prices with actual values, facilitating actionable insights and informed choices.

The rigorous comparison of the NN model's performance metrics (MAE, RMSE, SSE, R-squared) with those of other models (including regression trees, SVMs, and logistic regression) provides a holistic view of its effectiveness relative to alternative modeling techniques. This analysis assists in making well-informed decisions regarding model selection and deployment.

The selection of the NN model is underpinned by stringent criteria, including accuracy, generalizability, and the model's capacity to capture complex patterns. The acknowledgment of potential overfitting concerns, indicated by lower R-squared values on the test data, demonstrates a commitment to transparency and model evaluation. Furthermore, it's important to note that overfitting concerns can often be mitigated with additional regularization techniques or more extensive data.

Neural Network Results

The precise reporting of results, including specific metrics such as R-squared values of 0.9404 for training and 0.8734 for testing, as well as RMSE and MAE values, offers a comprehensive understanding of the model's performance. These metrics provide quantitative insights into predictive accuracy and error, enabling stakeholders to make informed decisions and set realistic expectations for the model's deployment.

The Neural Network model demonstrates impressive performance in predicting housing prices in King County, with substantial R-squared values of 0.9404 and 0.8734 for training and testing datasets, respectively. This indicates a high level of predictive accuracy, with the model explaining approximately 94% of the variance in the training dataset and 87% in the testing dataset. Such a high R-squared value in training suggests that the model fits the training data well, though the slight drop in the test data might hint at some overfitting.

The model's RMSE (Root Mean Squared Error) values provide insight into the average prediction error. With RMSE values of 94,836.04 for training and 150,299.41 for testing, we observe that the model's predictions are closer to the actual values in the training set compared to the test set. This again might indicate overfitting, where the model is tuned more closely to the training data's specific characteristics. The MAE (Mean Absolute Error) of 57,832.30 for training and 83,381.52 for testing reflects the model's average absolute error in predicting housing prices. The higher MAE in the test set compared to the training set further supports the possibility of overfitting.

The Regression Tree model

Regression trees are decision trees in which the target variables can take continuous values instead of class labels in leaves. Trees for numeric prediction are built in much the same way as they are for classification. Beginning at the root node, the data is partitioned using a divide and conquer strategy according to the feature that will result in the greatest increase in homogeneity in the outcome after a split is performed.

We built the regression tree model using the "rpart" method in the caret's train function. We also set the cross validation parameter to improve the model's performance. The Regression

Tree Model, a non-linear approach, shows R-squared values of 0.79117 for training and 0.75897 for testing. These figures, while lower than the linear models, still represent a decent level of predictive accuracy, especially considering the model's non-linear nature. The drop in R-squared from training to testing suggests some overfitting, typical in tree-based models.

The RMSE values stand at 165,274.84 for training and 186,382.77 for testing, indicating higher average prediction errors compared to the linear models. This could be due to the model capturing complex patterns in the training data that don't generalize as well to the test set. The MAE values of 109,635.94 for training and 116,844.59 for testing are the highest among all models discussed so far. This suggests that the average prediction error of the Regression Tree is comparatively larger, which might be a critical factor to consider in its practical application.

While the Regression Tree Model brings the advantage of modeling non-linear relationships, which are often present in real estate data, its tendency towards overfitting and higher error rates compared to linear models may limit its suitability as the primary model for predicting housing prices in King County.

VI. Model Limitations and Assumptions

Primary and Backup Models

The Neural Network model exhibited the highest R-squared values (0.9404 for training and 0.8734 for testing), indicating superior predictive accuracy. This model explained a significant portion of the variance in housing prices, more so than any other model tested. High R-squared in training shows the model’s ability to capture complex relationships in the data, while the relatively high value in testing indicates good generalization capabilities.

Although the RMSE and MAE values for the Neural Network model were higher in the testing set compared to the training set - suggesting some overfitting - these values were still among the lowest across all models. The RMSE of 150,299.41 and MAE of 83,381.52 in the testing set are indicative of relatively accurate predictions. Lower error metrics are crucial in real estate applications where large prediction errors can be costly.

Neural Networks are inherently more flexible and capable of modeling complex non-linear relationships that are often present in real estate data. This makes them well-suited for capturing intricate patterns and interactions among variables that linear models might miss.

Our backup for our primary model is the transformed model. The transformed model showed consistent R-squared values in both training (0.87256) and testing (0.87105), which were second only to the Neural Network model. This consistency is a strong indicator of the model's ability to generalize well, avoiding the pitfall of overfitting.

While the RMSE and MAE values of the Transformed Model were higher compared to the Neural Network, especially in the testing set (RMSE of 200,676.10 and MAE of 81,099.88), they were still within a reasonable range. These figures suggest that the model is capable of making relatively accurate predictions, albeit not as precise as the Neural Network.

The 'Transformed Model' involves some form of data transformation or feature engineering. It implies that it effectively handles outliers and skewed distributions, which are common in housing data. Such transformations can make the model robust to anomalies in the data, which is a valuable quality in real estate market analysis.

While the Neural Network showed some signs of overfitting (as evidenced by the drop in R-squared from training to testing), its overall performance metrics still outperformed other

models. The Transformed Model, with its consistent performance across training and testing, serves as a good backup to mitigate any overfitting risks.

In real estate price prediction, minimizing prediction errors is crucial due to the high stakes involved. Both the Neural Network and the Transformed Model presented reasonable error metrics, making them preferable for such applications. While simpler models like OLS Linear or OLS Stepwise models offer greater interpretability, they fell short in capturing the complexity of the housing market data as effectively as the Neural Network. The Transformed Model strikes a balance between complexity and interpretability, making it a suitable backup.

Overall, the Neural Network model's superior ability to capture complex patterns and its lower error metrics make it the best choice for predicting housing prices in King County. The Transformed Model, with its consistent performance and potential robustness, serves as an excellent backup, especially in scenarios where overfitting or complexity of the primary model might be a concern.

Benchmark

We have decided to choose the Lasso Model as the benchmark for predicting housing prices in King County, several key factors have been considered. This analysis assesses the Lasso Model's performance with regard to linearity, normality, multicollinearity, outliers, and heteroscedasticity. The examination primarily focuses on the model's coefficients and their differences (total_diff) in comparison to other robust models (Huber, Ridge, Elastic Net, and Lasso).

The Lasso Model demonstrates a relatively low total_diff value compared to alternative models. This metric serves as a significant indicator of the model's overall stability and

consistency in estimating coefficients across different modeling techniques. Lasso inherently addresses multicollinearity by shrinking certain coefficients to zero. This feature selection technique enhances model interpretability, reduces overfitting, and effectively deals with highly correlated predictors.

Lasso's regularization method not only facilitates feature selection but also mitigates sensitivity to outliers. This is evident in the stability of coefficients associated with variables such as 'lat,' 'long,' 'sqft_living,' and 'grade,' which are critical in housing price predictions. By penalizing the absolute magnitude of coefficients, Lasso diminishes the influence of outliers and aids in handling heteroscedasticity.

The housing market dataset often includes a mix of categorical (e.g., zipcode, condition, grade) and continuous variables (e.g., sqft_living, bedrooms). Lasso adeptly manages this complexity by selecting the most relevant features and mitigating overfitting, as indicated by its coefficients across various predictors. In comparison to models such as Ridge or Elastic Net, Lasso strikes a balance between coefficient shrinkage and feature selection. Unlike Ridge, which shrinks coefficients but retains all variables, Lasso's approach of setting some coefficients to zero simplifies the model and enhances interpretability.

The Lasso Model's coefficients align with intuitive real estate dynamics. For instance, positive coefficients for 'sqft_living,' 'bathrooms,' and 'grade_Above_Average' correspond to the understanding that larger, higher-graded houses with more amenities generally command higher prices. Negative coefficients for certain zip codes may suggest less desirable areas, a common insight in real estate analysis. While Lasso does not directly address linearity or normality assumptions, its reduction of multicollinearity indirectly contributes to a more stable and reliable linear regression framework.

Limitations

When considering limitations for our model choices, for our transformed model, we see that we improved the linearity of the OLS model with the lambda transformation. Nevertheless we didn't solve for unequal variances and normality of residuals as there are several outlying points and removing them may reduce the model's performance. This requires monitoring as the model goes into use. This same situation applies to our benchmark model. Lasso solves for multicollinearity but it can be sensitive to outliers.

Finally, a major limitation of working with NN is that it is hard to interpret neural networks. We can have some performance and diagnostic metrics but we cannot know exactly what is happening inside the model.

VII. Ongoing Model Monitoring

It's very important to put in place an ongoing model monitoring to guarantee the model's performance and stability. Our plan is to assess on a daily basis the performance and stability of our Best, Backup and Benchmark models. We aim to add thresholds and an alert system in terms of the difference between the current R-squared on test dataset and the R-squared on test dataset originally calculated as well as the difference between the current R-squared on the test data set and the current R-squared on the training dataset. We can also use our benchmark model as the minimum for performance.

If these thresholds are surpassed, we receive an alert to then intervene. Once we have identified the issue, we can take steps to do the following: replace the best model with the backup model while we investigate it. Following the investigation, replace the benchmark model if necessary.

Our final goal is to always have the best model with high performance on the test data set and high stability between training and test data. Moreover, it's always important to have a backup model in the case of an instability, and a benchmark model as our minimum acceptable performance, stability and linearity assumptions.

VIII. Conclusion

The Lasso Model's effective control of multicollinearity, handling of outliers, and balanced approach to feature selection and regularization make it a well-suited benchmark model for predicting housing prices. Its lower total_diff value and the interpretability of its coefficients, especially within the context of real estate data, further support its appropriateness for this application. This approach, while robust, maintains a crucial level of simplicity and interpretability that is essential for practical real estate market analyses.

The Neural Networks model is our best model and our primary choice for the following reasons: it has the highest R-squared values, balanced RMSE and MAE, and the model's strong flexibility and complexity. In addition, we selected a backup model in case the Neural Network model loses performance or stability. Our backup model is the Transformed Model as it has consistent R-squared across datasets, reasonable RMSE and MAE and potential for robustness in feature handling.

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Appendix: Please refer to HTML for full tables and sections.

Figure 1.1 Data Description

Variable	Description
id	Unique ID for each home sold (not used as a predictor)
date	Date of the home sale
price	Price of each home sold
bedrooms	Number of bedrooms
bathrooms	Number of bathrooms, ".5" accounts for a bathroom with a toilet but no shower
sqft_living	Square footage of the apartment interior living space
sqft_lot	Square footage of the land space
floors	Number of floors
waterfront	A dummy variable for whether the apartment was overlooking the waterfront ...
view	An index from 0 to 4 of how good the view of the property was
condition	An index from 1 to 5 on the condition of the apartment
grade	An index from 1 to 13 about building construction and design quality
sqft_above	The square footage of the interior housing space above ground level
sqft_basement	The square footage of the interior housing space below ground level
yr_built	The year the house was initially built
yr_renovated	The year of the house's last renovation
zipcode	The zipcode area the house is in
lat	Latitude coordinate
long	Longitude coordinate
sqft_living15	The square footage of interior housing living space for the nearest 15 neighb...
sqft_lot15	The square footage of the land lots of the nearest 15 neighbors

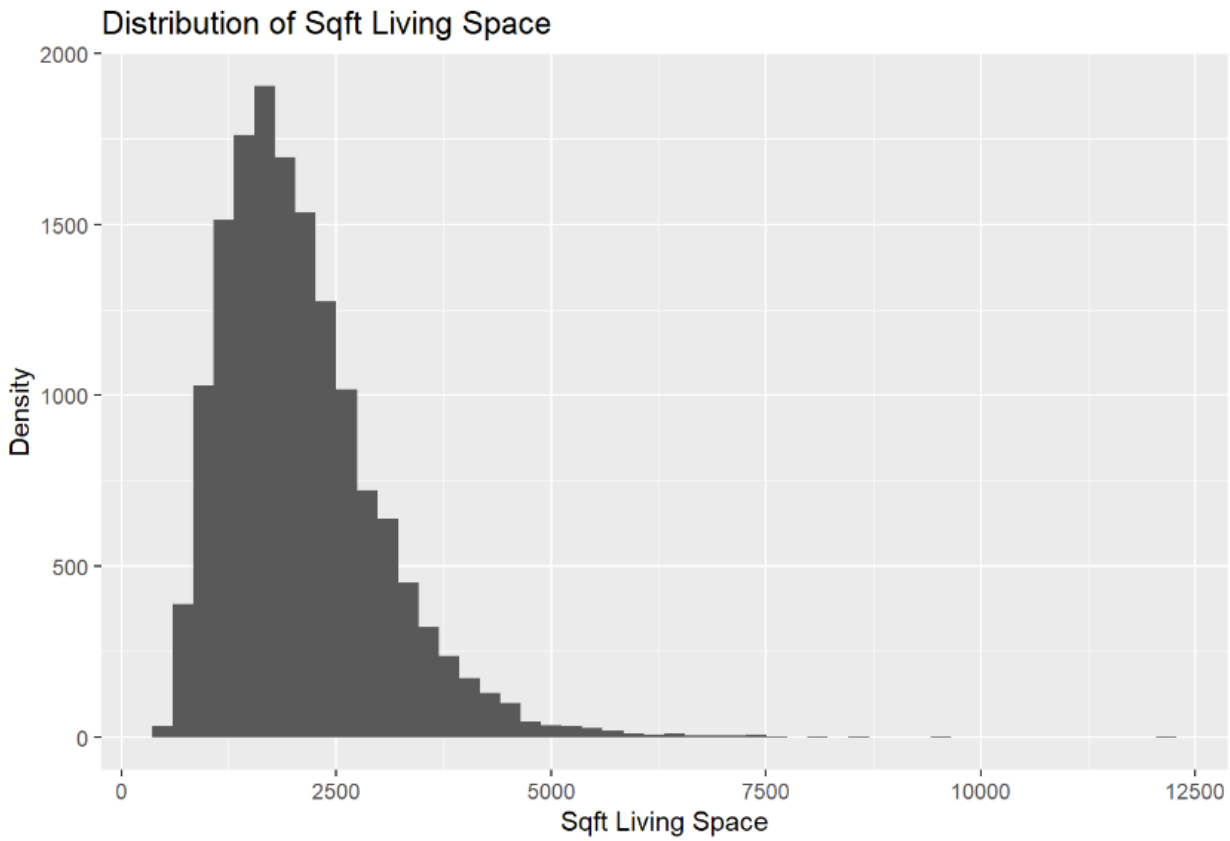

Figure 5.2 Distribution of Sqft Living Space

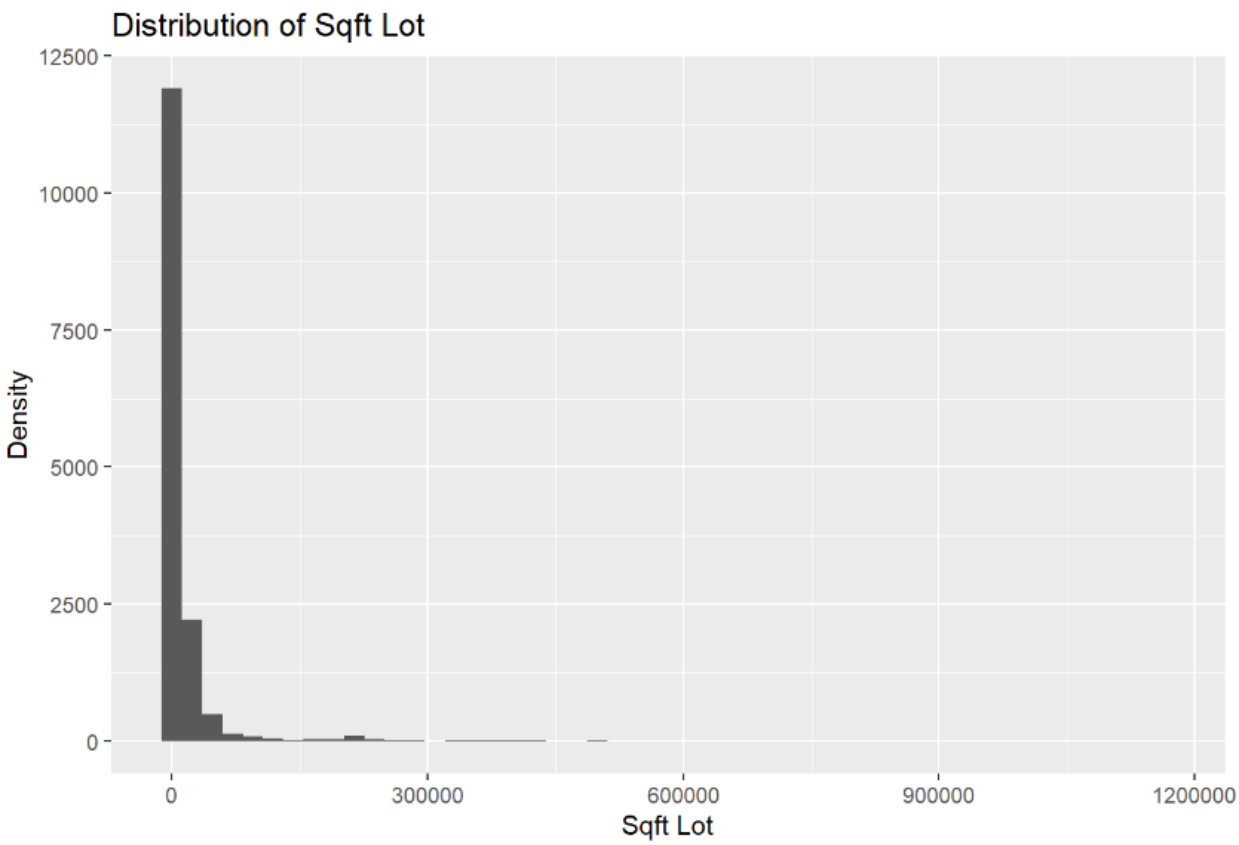

Figure 5.4 Distribution of Sqft Lot

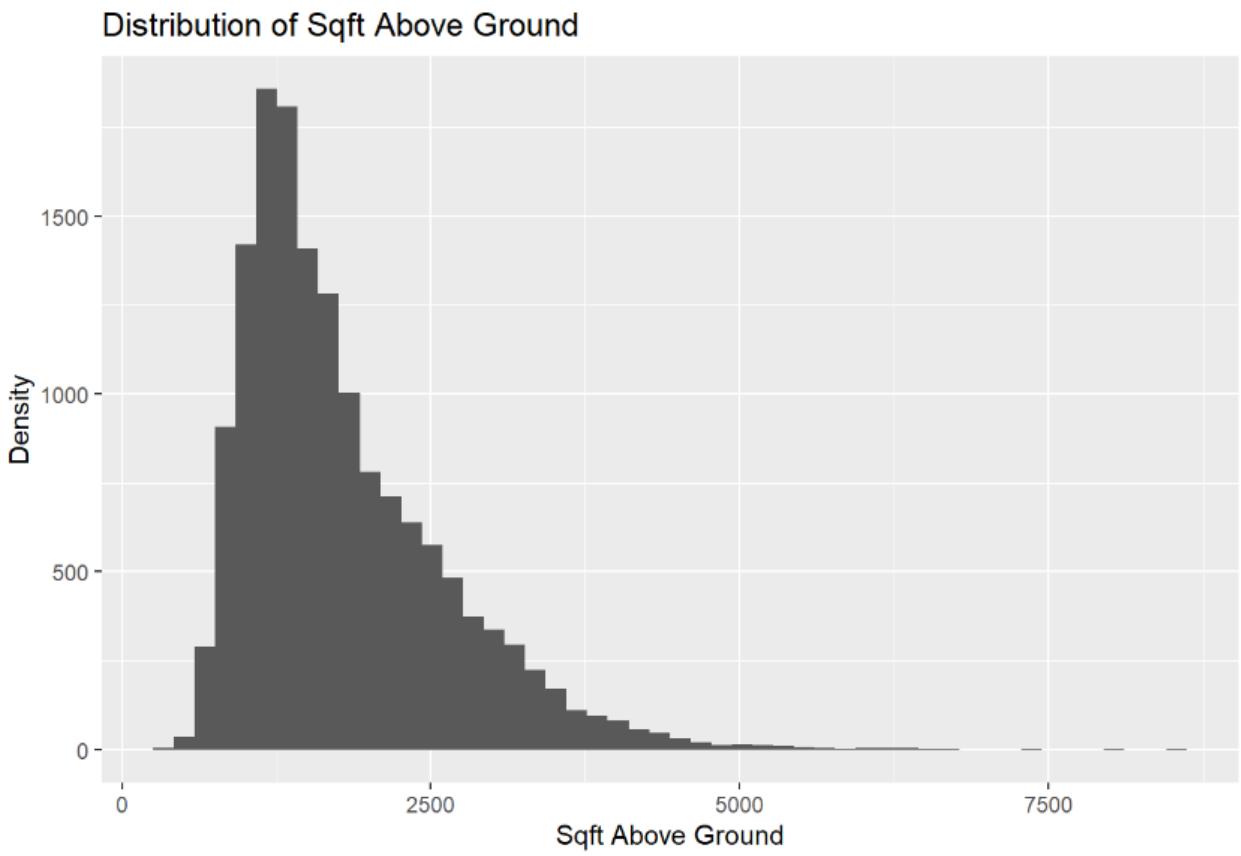

Figure 5.6 Distribution of Sqft Above Ground

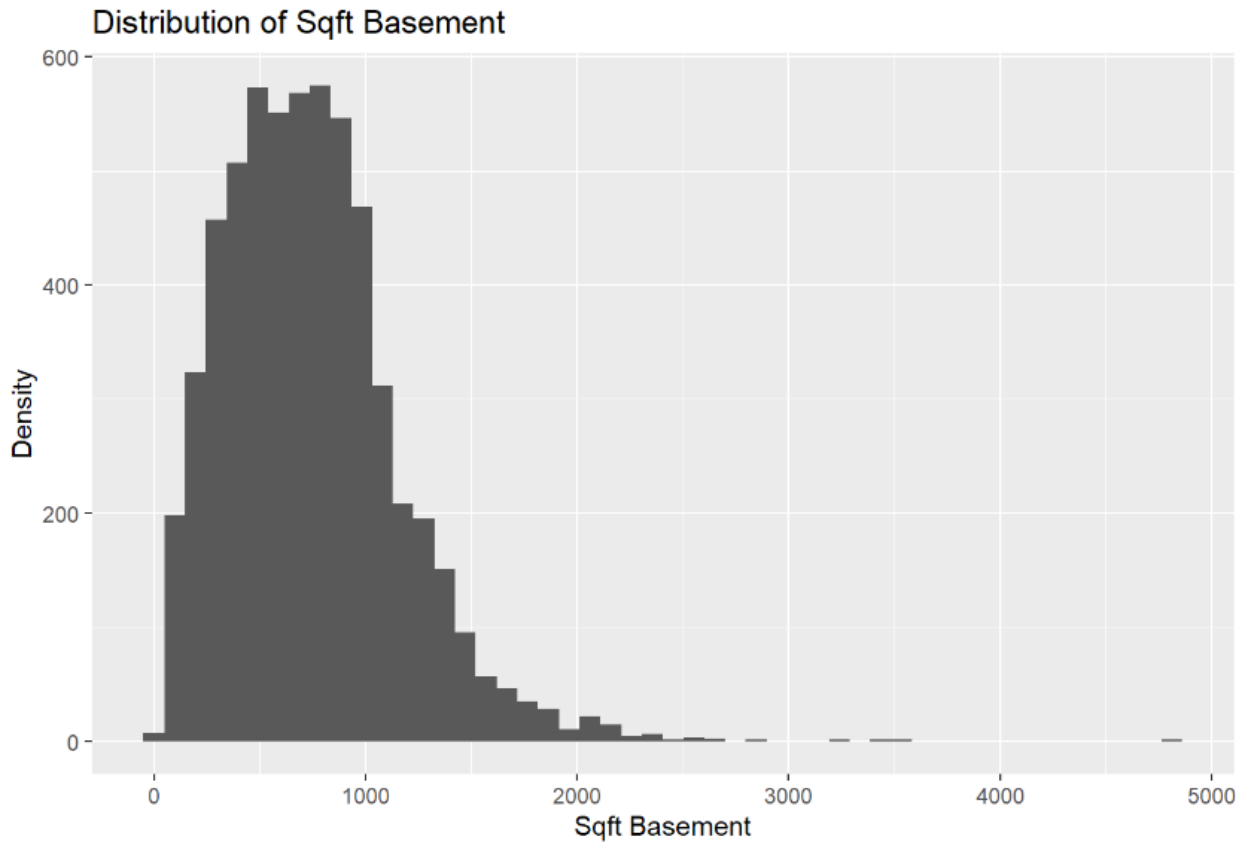

Figure 5.8 Distribution of Sqft Basement (Non-Zero Values)

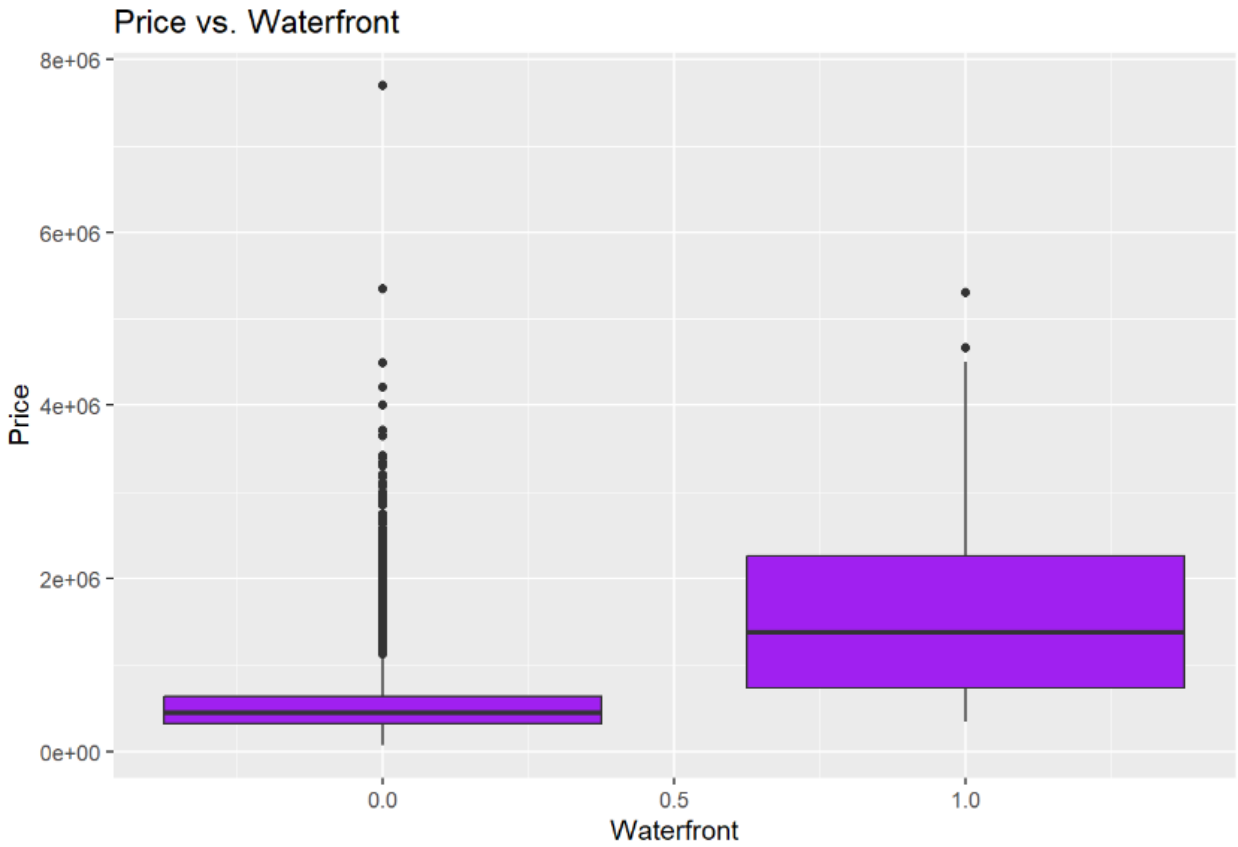

Figure 5.21 Price vs. Waterfront

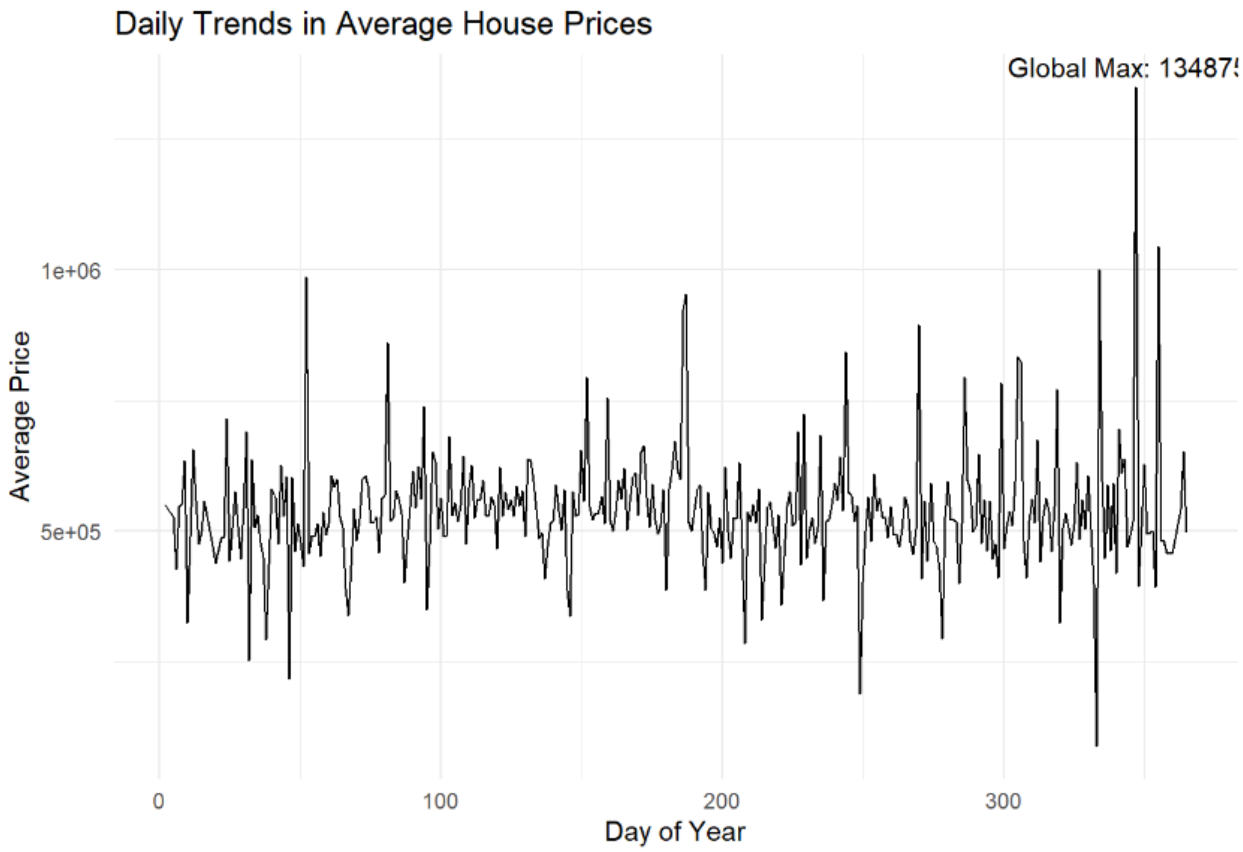

Figure 5.36 Daily Trends in Average House Prices

Figure 5.38 Distance to Convergence Map

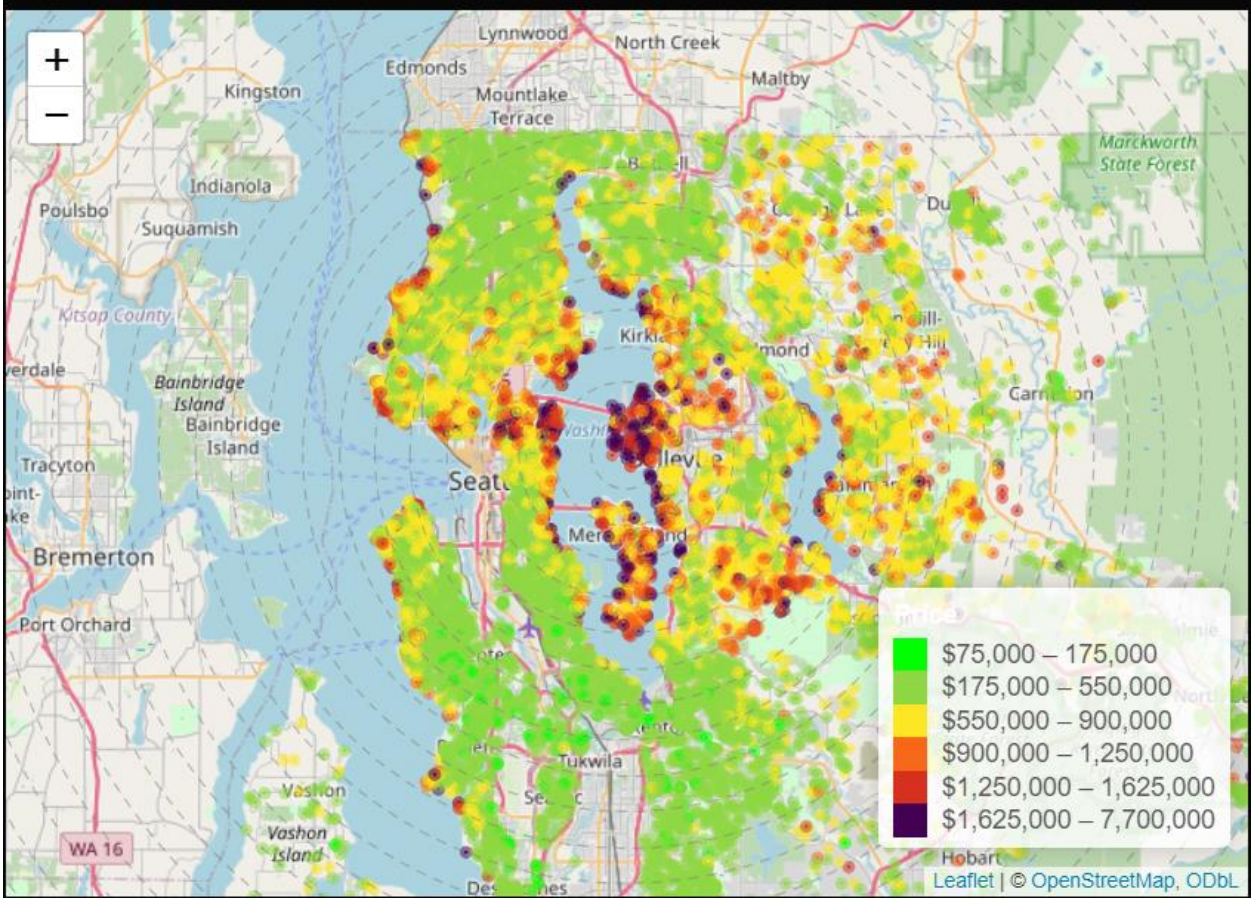

Table 6.1 Dropped Features

Stepwise Method	Dropped Features
OLS_Step_Backward	yr_built, sqft_lot15, condition_2, grade_Average, season_Summer
OLS_Step_Forward	sqft_lot15, day_sold, condition_2, season_Fall
OLS_Step_Both	yr_built, sqft_lot15, day_sold, zipcode_98010, zipcode_98022, zipc…

Table 6.2 Step Model Additions

	Model	SSE_train	SSE_test	R_squared_train	R_squared_test	RMSE_train
1	OLS_linear	358990638973129.3	168418665362960.9	0.81859	0.81966	154041.04
2	OLS_Step_Both	363845493464973.4	170765689586178.3	0.81614	0.81715	155079.14
3	OLS_Step_Backward	360535123293971.4	168396827583548.8	0.81781	0.81969	154372.05
4	OLS_Step_Forward	360549949155043.8	168472479791881.8	0.8178	0.81961	154375.23

Figure 7.1 Residuals vs Fitted Plot for Linear Model

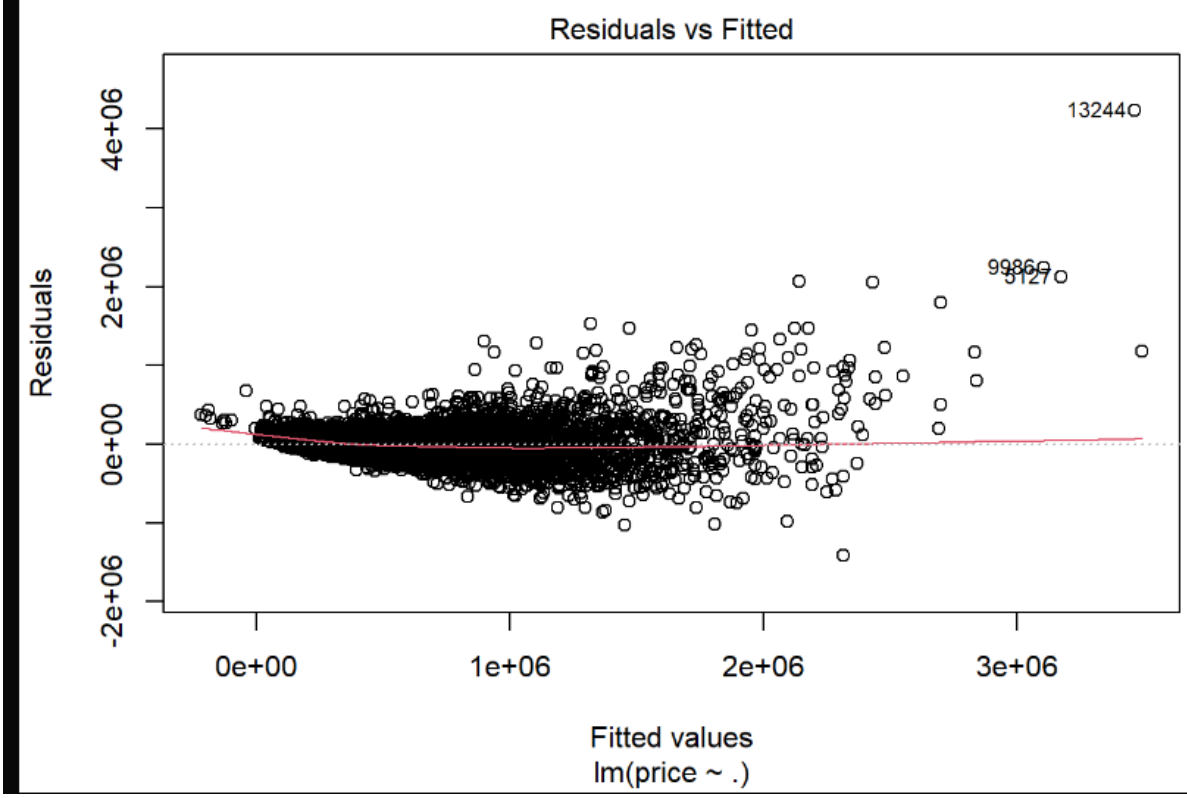

Figure 7.4 QQ Plot for Linear Model

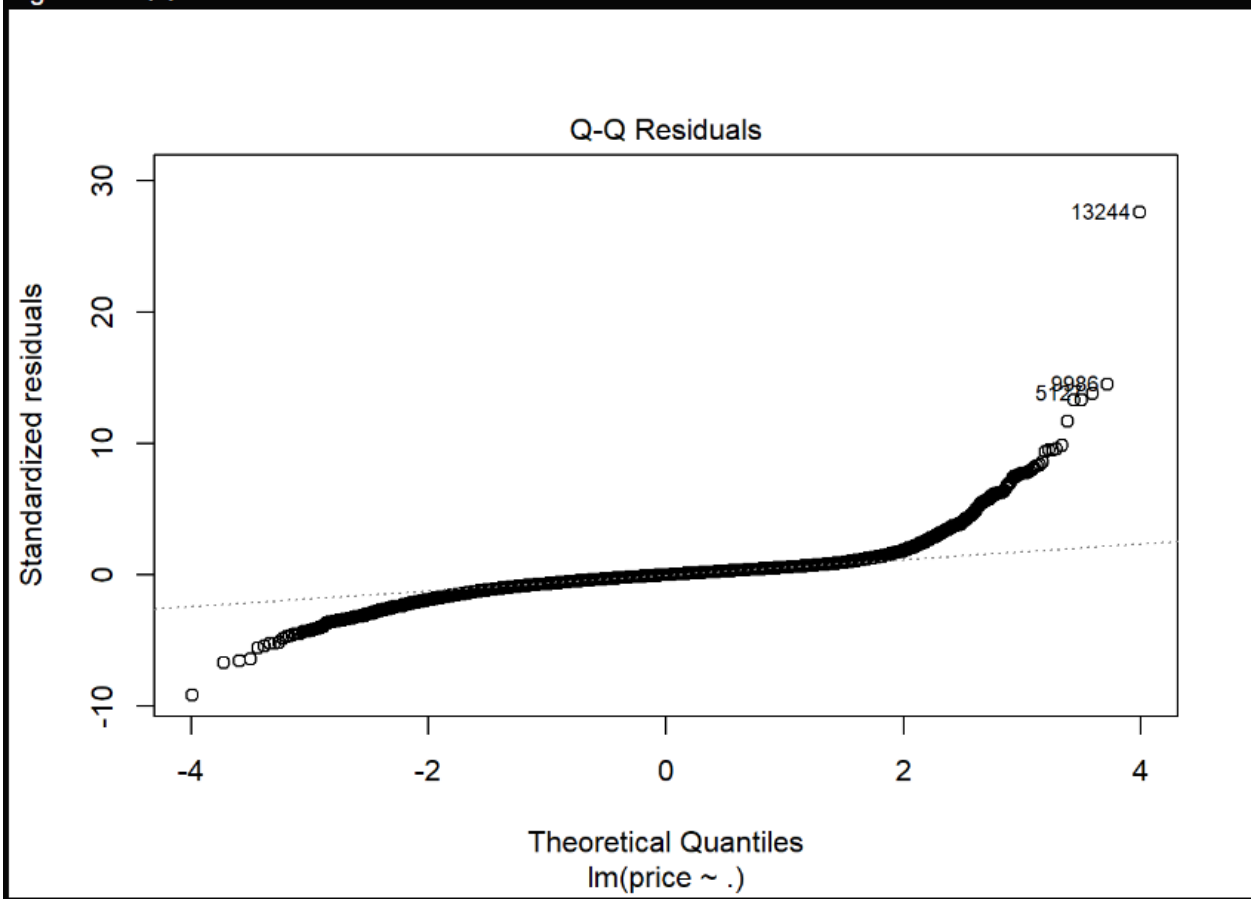

Figure 7.5 Scale-Location Plot for Linear Model

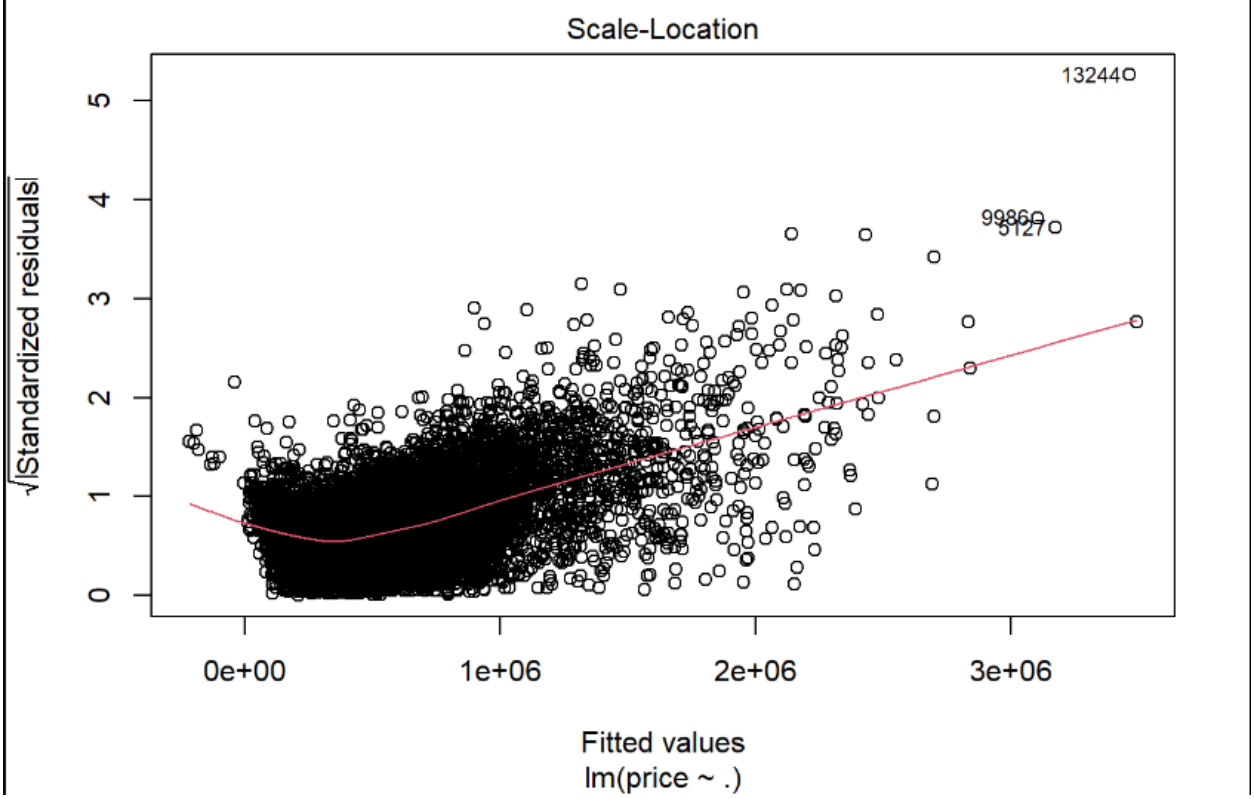

Figure 7.6 Cooks Distance Plot for Linear Model

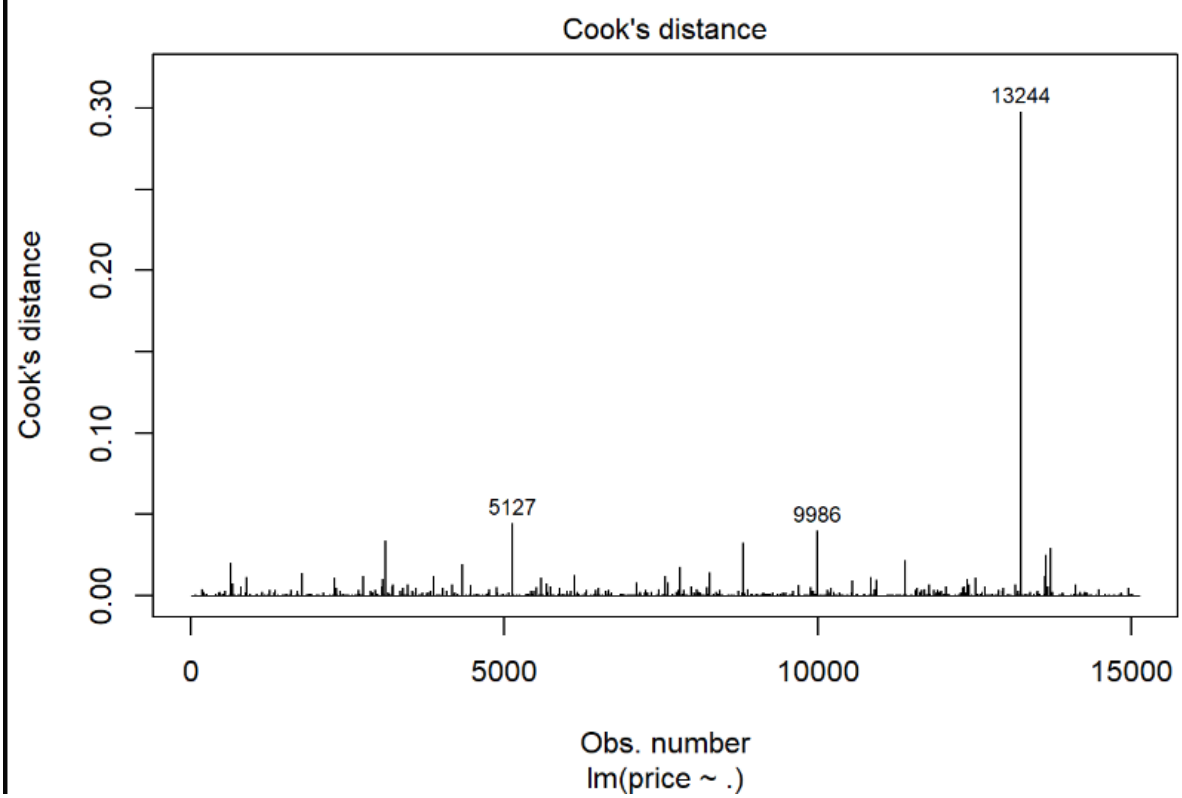

Figure 8.1 Box-Cox Plot for Linear Model

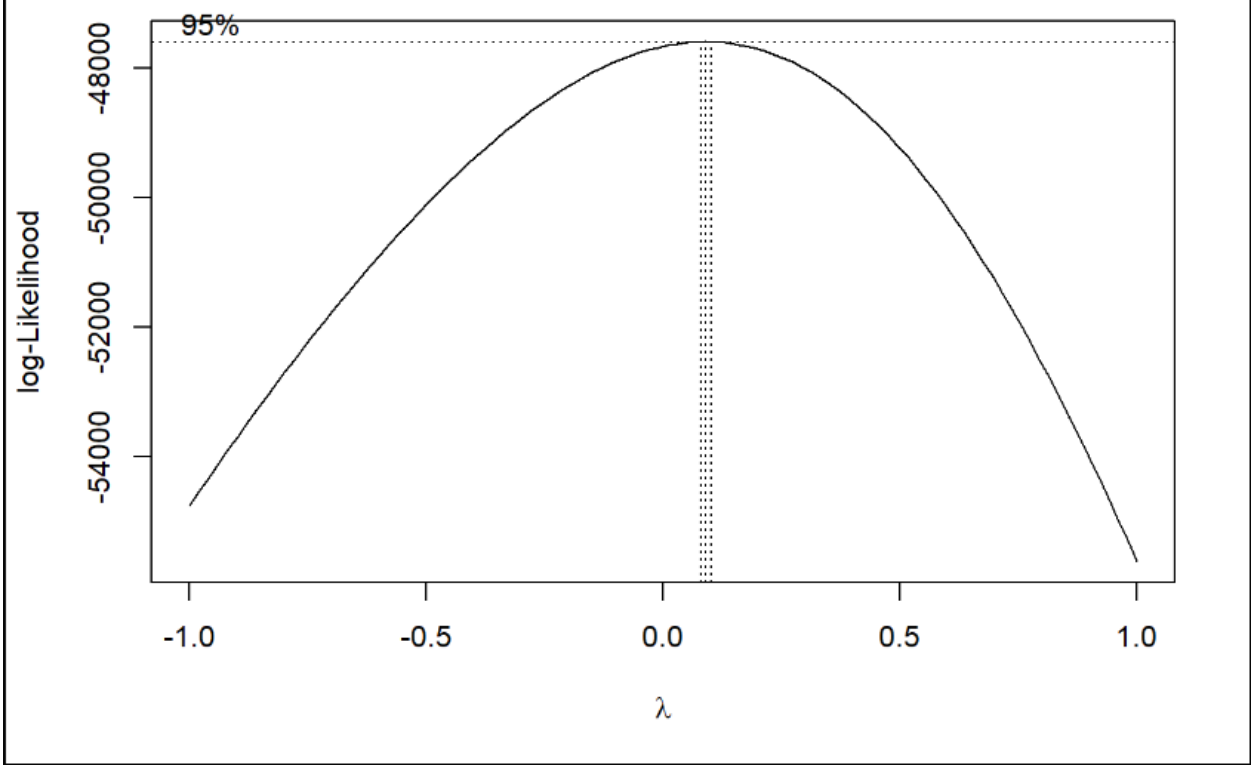

Figure 8.2 Diagnostic Plots after Transformation

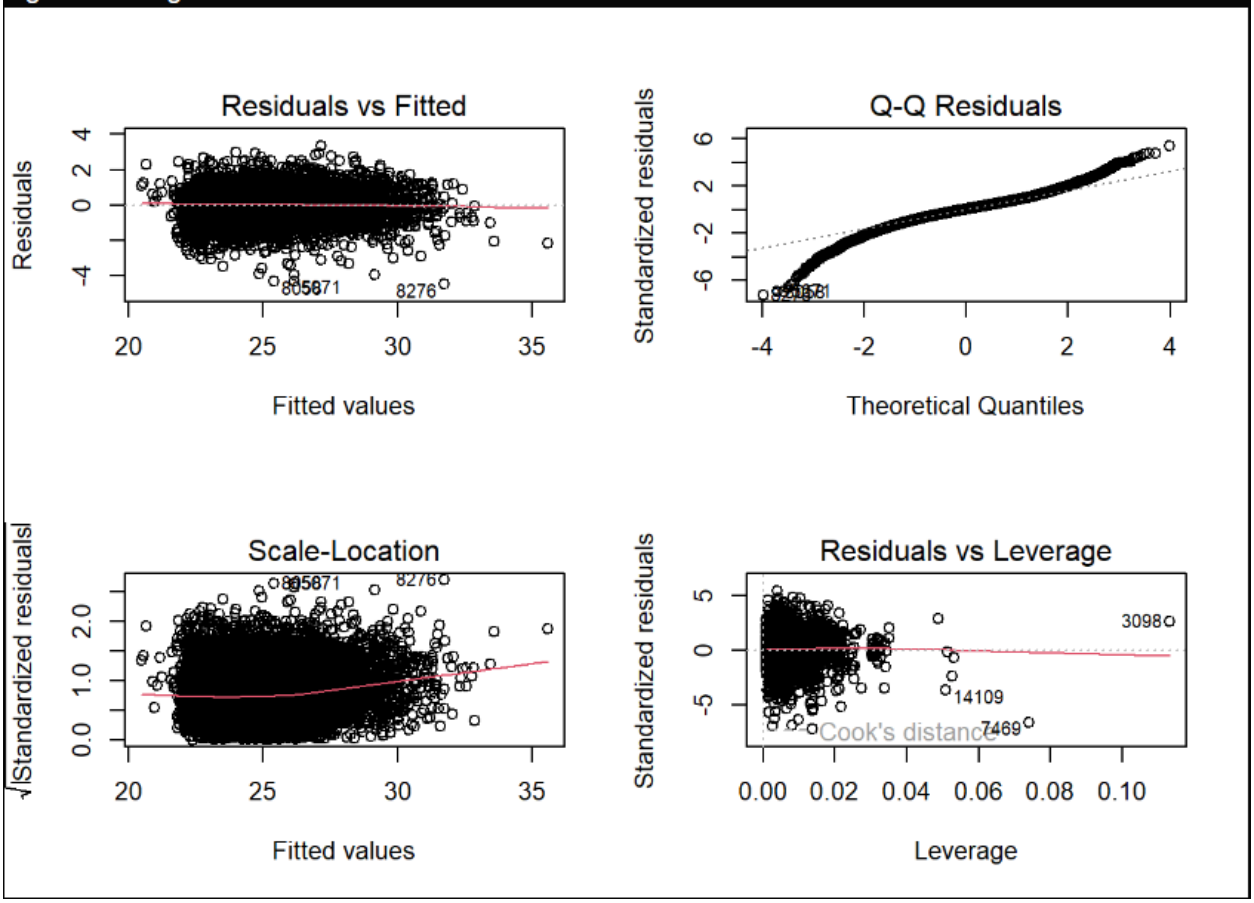

Figure 7.7 OLS Cooks Distance Plot for Linear Model

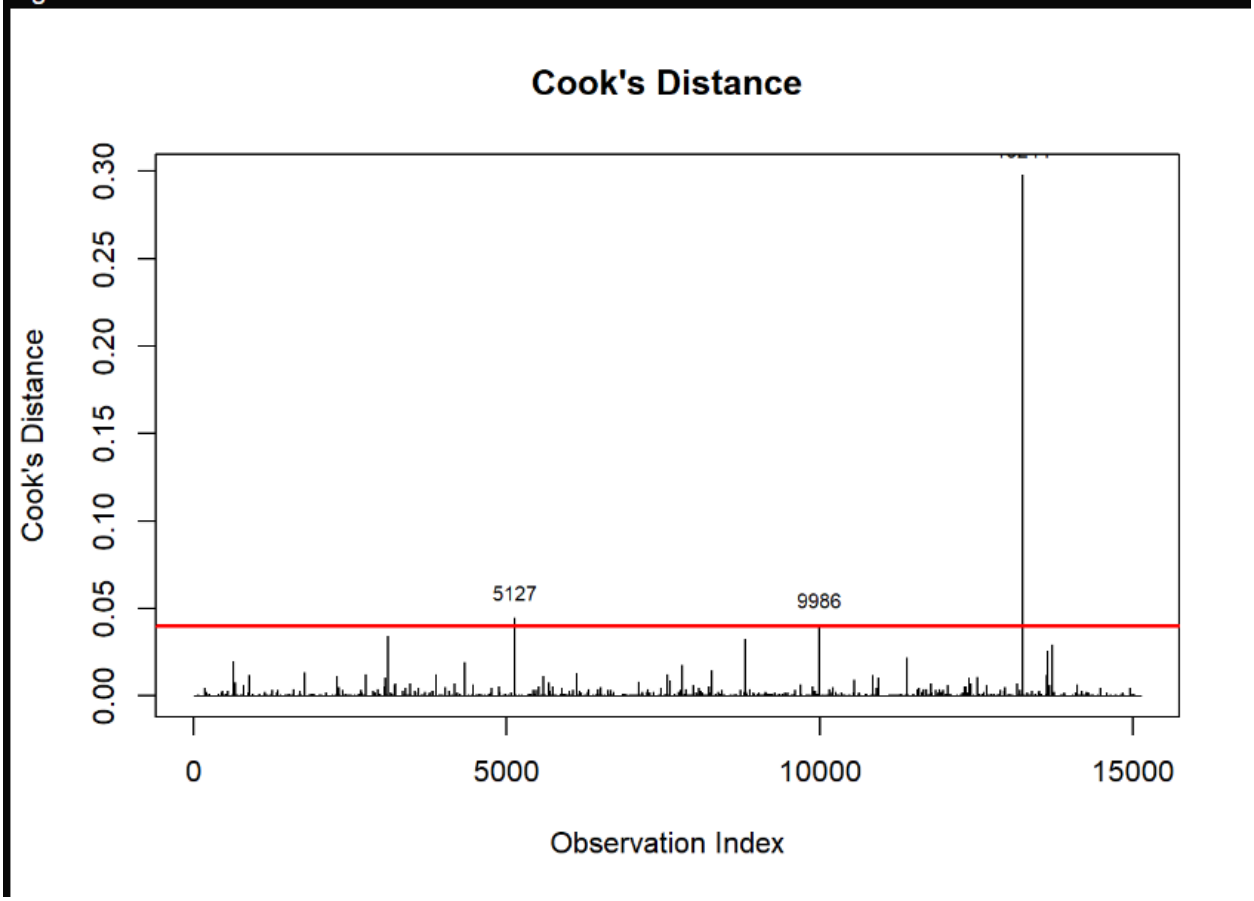

Table 8.2 Weighted Transformation Model Addition

	Model	SSE_train	SSE_test	R_squared_train	R_squared_test	RMSE_trai
1	OLS_linear	358990638973129.3	168418665362960.9	0.81859	0.81966	154041
2	OLS_Step_Both	363845493464973.4	170765689586178.3	0.81614	0.81715	155079
3	OLS_Step_Backward	360535123293971.4	168396827583548.8	0.81781	0.81969	154372
4	OLS_Step_Forward	360549949155043.8	168472479791881.8	0.8178	0.81961	154375
5	OLS_Step_Both_Outl…	335401461156400.6	171189574711398	0.82176	0.8167	148908
6	Transformed Model	268306309736922.8	261116484128967.5	0.87256	0.87105	133184
7	WT Model 200	6253102748259004	2847613794116864	0.75884	0.74016	642962